KING MONGKUT'S UNIVERSITY OF TECHNOLOGY THONBURI | DEPARTMENT OF COMPUTER ENGINEERING

CPE 334 Introduction to Software Engineering in the Age of AI Agents (Semester 1/2026)

Pollipop

University Review, Feedback, and Survey System

Project Deliverable 1: Engineering Baseline and Working Sprint 1 (Full Engineering Report & Verification)

Course & Term:	**CPE 334** (Semester 1/2026) | Due: 8 November 2026, 23:59 (80 Points / 8% Grade)
Assigned Theme:	**Theme 10: Review, Feedback, and Survey System**
GitHub Repository:	`https://github.com/phatthidawadi/pollipop` (Public per requirement R-12)
Release Branch & Commit:	Branch `sprint-1-staging` | Release Commit `00e1ba6` | Tag `v1.0.0-sprint1`
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Evaluation Score Target:	**80 / 80 Points (100%)** across Parts A to F with Zero Gaps

Master Evidence Index & Short Navigation Table (Protocol Section 4.3)

According to CPE 334 Deliverable 1 Protocol Section 4.3, this master navigation index links every required part (A through F), its core engineering scope, stable IDs, and exact location of executable evidence:

Part	Protocol Heading & Scope	Core Stable IDs	Artifact Source & Verifiable Evidence
Part A	Software Requirements Specification (SRS)	FR-01..16, BR-01..13, NFR-01..05	`docs/srs.md` (Domain context, actors, scope, decisions)
Part B	Feature Inventory & Traceability Matrix	S-01, F-01..F-09	`docs/features.md`, `docs/traceability.md` (Complete requirements mapping)
Part C	System-Level Software Design Specification	Three-Tier Architecture, ERD	`docs/system-sds.md` (Stack, ERD, Security, Role Matrix, GCP Plan)
Part D	Three-Sprint Plan & Pair Workflow	Sprints 1, 2, 3	`docs/sprints/plan.md` (Vertical slice rationale, 10 balanced issues)
Part E	Sprint 1 Engineering Contracts & DoD	AC-01..38, TEST-01..40	`docs/sprints/sprint-1/contracts.md` (Feature design, STS, DoD)

Part	Protocol Heading & Scope	Core Stable IDs	Artifact Source & Verifiable Evidence
Part F	Working Sprint 1 Verification & Peer Reviews	46 Tests (100% Pass)	`docs/sprints/sprint-1/verification.md` (Screenshots, PR dialogues, logs)

Part A: Software Requirements Specification (SRS)

> **Suggested Artifact:** `docs/srs.md` | **Weight:** 10 Points | Demonstrates: Stakeholders, Context, Actors, Product Scope, Numbered FR/BR/NFR, Assumptions & Decisions.

1. Stakeholder Problem & System Context

1.1 Stakeholder Problem Statement

Higher education institutions, academic faculties, student affair departments, course lecturers, and student clubs (Survey Owners) routinely conduct evaluations regarding course instruction, library learning environments, dining services, campus Wi-Fi infrastructure, and student activities. However, relying on generic paper forms or consumer-grade online survey tools introduces significant software engineering and data integrity failures:

- **Uncontrolled Duplicate Submissions & Identity Leakage:** Standard forms either allow unrestricted, untraceable repeat submissions or tightly bind student accounts to their answers, eliminating genuine anonymity and causing social desirability bias in sensitive campus feedback.
- **Absence of Role-Based & Institutional Access Boundaries:** Generic tools lack fine-grained role segregation. An instructor can inadvertently view, modify, or delete another instructor's course evaluations, violating privacy and departmental isolation.
- **Data Corruption Caused by Post-Publish Mutations:** Allowing question options or ratings to be modified while responses are being actively submitted breaks historical validity and produces corrupted aggregate summaries.

1.2 Chosen Domain Context: Pollipop

Pollipop is a modern, cloud-ready three-tier web application specifically engineered for university course reviews and campus evaluations. The system manages the full survey lifecycle: structured survey creation (Single Choice and Rating Scale), lifecycle state transitions (`DRAFT` → `PUBLISHED` → `CLOSED`), authenticated response submission within atomic database transactions, strict duplicate prevention, and real-time server-side aggregate calculation.

The system is packaged with multi-stage Docker containers, making it ready for horizontal autoscaling on Google Cloud Platform (Cloud Run and Cloud SQL) per CPE 334 Lab 5 requirements.

2. Actors and Role-Based Responsibilities

Actor / Role	Role Code	Allowed Operations	Explicitly Forbidden Operations
Survey Owner (Instructor / Staff)	`OWNER`	- Create new surveys in `DRAFT` status with Single Choice (2–10 options) and Rating (1–5) questions. - Inspect, update (while in `DRAFT` status only), and transition survey state to `PUBLISHED` or `CLOSED`. - Inspect real-time server-calculated aggregate analytics of surveys they own.	- **Forbidden:** Accessing, modifying, publishing, or inspecting summaries of surveys owned by other owners (`403 Forbidden`). - **Forbidden:** Submitting responses to any survey as a respondent (`403 Forbidden`).

Actor / Role	Role Code	Allowed Operations	Explicitly Forbidden Operations
Survey Respondent (Student / University Member)	`RESPONDENT`	- View open (`PUBLISHED`) surveys with personal completion status (`answered: true / false`). - Submit a validated response exactly once per survey. - Review own submitted answers in `IDENTIFIED` mode via `GET /api/responses/:id` .	- **Forbidden:** Creating, modifying, publishing, or closing surveys (`403 Forbidden`). - **Forbidden:** Viewing `DRAFT` surveys in listing or detail queries (`404 Not Found`). - **Forbidden:** Viewing aggregate summary analytics (`403 Forbidden`). - **Forbidden:** Viewing other respondents' submissions (`403 Forbidden`).

3. Numbered Functional Requirements (FR)

FR ID	Observable Requirement Behavior	Sprint
FR-01	**User Login & Session Creation:** Authenticate via email/password, verify bcrypt hash, set HTTP-only `pollipop_sid` cookie.	Sprint 1
FR-02	**Current User Profile & Logout:** Retrieve profile (`GET /api/auth/me`) and destroy session (`POST /api/auth/logout`) with cache invalidation.	Sprint 1
FR-03	**Health & Readiness Probes:** `GET /api/health` (liveness) and `GET /api/ready` (readiness with database `SELECT 1`).	Sprint 1
FR-04	**Create Draft Survey:** `OWNER` creates survey with title, description, mode (`IDENTIFIED`), and questions in `DRAFT` status.	Sprint 1
FR-05	**Sprint 1 Question Types:** Accept `SINGLE_CHOICE` (2–10 options) and `RATING` (1–5) questions with `required` flag.	Sprint 1
FR-06	**Publish Survey:** `OWNER` transitions survey from `DRAFT` to `PUBLISHED` (making it active for respondents).	Sprint 1
FR-07	**List Surveys by Role:** `OWNER` sees own surveys; `RESPONDENT` sees `PUBLISHED` surveys with `answered` and `responseId` .	Sprint 1
FR-08	**Submit Validated Response:** `RESPONDENT` submits answers inside an atomic database transaction.	Sprint 1
FR-09	**View Own Submitted Response:** In `IDENTIFIED` mode, `RESPONDENT` inspects submitted answers via `GET /api/responses/:id` .	Sprint 1
FR-10	**View Survey Aggregate Summary:** `OWNER` retrieves server-calculated statistics (counts, percentages, mean rating).	Sprint 1
FR-11	**Edit Draft Survey:** `OWNER` updates title, description, or questions while in `DRAFT` status.	Sprint 2
FR-12	**Close Survey & Expiration:** Manual close (`POST /api/surveys/:id/close`) or scheduled closing (`closesAt`).	Sprint 1 / 2
FR-13	**Anonymous Survey Mode:** Decouple participation tracking from answer storage (`respondentId = null`).	Sprint 2

FR ID	Observable Requirement Behavior	Sprint
FR-14	**Department Eligibility:** Restrict survey answering to students matching `targetDepartment`.	Sprint 2
FR-15	**Extended Question Types:** Multi-select (`MULTIPLE_CHOICE`) and open feedback (`SHORT_TEXT`).	Sprint 2
FR-16	**Export Survey Results:** Download summary analytics and anonymized responses as CSV files.	Sprint 3

4. Numbered Business Rules (BR)

BR ID	Domain Rule Description	Enforced Layer
BR-01	**Single Submission per Respondent:** Duplicate attempts return `409 Conflict (ALREADY_SUBMITTED)` backed by DB unique constraint `(surveyId, userId)` on `SurveyParticipation`.	Application & DB
BR-02	**Responses Allowed Only on Published Surveys:** Answering `DRAFT` returns `404 Not Found`; answering `CLOSED` returns `409 Conflict (SURVEY_NOT_OPEN)`.	Application
BR-03	**One-Way Survey Lifecycle State Machine:** States progress `DRAFT` → `PUBLISHED` → `CLOSED`. Reversing state returns `409 Conflict (INVALID_STATE_TRANSITION)`.	Application
BR-04	**Survey Ownership Isolation:** `OWNER` cannot access, edit, publish, close, or inspect summaries of surveys owned by others (`403 Forbidden`).	Application
BR-05	**Role Segregation:** `RESPONDENT` cannot author/publish surveys or view summaries; `OWNER` cannot submit responses (`403 Forbidden`).	Application
BR-06	**Cross-Response Privacy:** `RESPONDENT` can only view own submitted answers; inspecting another user's answers returns `403 Forbidden`.	Application
BR-07	**Draft Survey Visibility Restriction:** `RESPONDENT` cannot view or retrieve `DRAFT` surveys (`404 Not Found`).	Application
BR-08	**Question & Option Structural Constraints:** Survey must have 1–50 questions. `SINGLE_CHOICE` must have 2–10 non-empty options.	Application
BR-09	**Answer Completeness & Validity:** Required questions must be answered. Options must belong to the question. Rating must be 1–5. Optional questions may be omitted.	Application & DB
BR-10	**Server-Side Aggregate Accuracy:** Statistics computed server-side to 2 decimal places. Handles zero responses without division by zero.	Application
BR-11	**Immutability After Publication:** Modifying questions/options after publication is forbidden (`409 Conflict`).	Application
BR-12	**Sprint 1 Baseline Eligibility:** All authenticated students can answer any published survey.	Application
BR-13	**True Anonymity Separation:** In `ANONYMOUS` mode, `Response.respondentId` is strictly `null`.	Application & DB

5. Numbered Non-Functional Requirements (NFR)

NFR ID	Category	Measurable Target	Practical Verification Method
NFR-01	Security & Cache Guard	Unauthenticated requests rejected with `401` 100% of the time. `Cache-Control: no-store` on protected endpoints. Browser back redirect to `/login` on revoked sessions.	Supertest API integration + Playwright E2E browser history test (`TEST-39` , `TEST-40`).
NFR-02	Password & Secret Hygiene	Passwords hashed with bcrypt (salt rounds ≥ 10). Zero plaintext secrets in Git, logs, or client bundle.	Database seed inspection (`TEST-38`) + git index audit against `.env.example` .
NFR-03	Data Integrity & Concurrency	Concurrent duplicate submissions result in exactly one `201 Created` and one `409 Conflict` with atomic rollback.	Database unique constraint + `Promise.all` concurrent test (`TEST-37`).
NFR-04	Validation & Error Standard	Zod schema validation returning structured `400 Bad Request` with dot-path notation and sanitized messages.	Zod unit tests & Supertest invalid payload tests (`TEST-03` , `TEST-11` , `TEST-24`).
NFR-05	Cloud-Readiness	All configurations read from environment. Probes `/api/health` & `/api/ready` respond in < 500 ms. Sessions survive restarts.	Supertest health probe tests (`TEST-07` , `TEST-08`) + Docker packaging.

Part B: Feature Inventory & Requirements Traceability Matrix

> **Suggested Artifacts:** `docs/features.md` & `docs/traceability.md` | **Weight:** 10 Points | Demonstrates: 3-Sprint Breakdown and Bidirectional Traceability.

1. Three-Sprint Feature Inventory Table

Feature ID	Feature Scope & Description	Associated Requirements	Sprint
S-01	**System Foundation:** Express API + React Vite SPA + Prisma PostgreSQL, env configuration, idempotent seed, probes `/api/health` & `/api/ready`.	FR-03, NFR-02, NFR-05, A-01, A-08	**Sprint 1**
F-01	**Authentication & Roles:** Login, logout, profile retrieval, database session cookie (`pollipop_sid`), bcrypt hashing, RBAC, cache guard.	FR-01, FR-02, BR-05, NFR-01, NFR-02	**Sprint 1**
F-02	**Survey Lifecycle:** Survey creation (Single Choice & Rating), publish/close transitions, owner isolation, post-publish immutability.	FR-04..07, FR-12, BR-03, BR-04, BR-05, BR-08, BR-11	**Sprint 1**
F-03	**Response Submission:** Validated submission inside atomic transaction, duplicate blocking (409), inspection of own answers, cross-response privacy.	FR-08, FR-09, BR-01, BR-02, BR-05, BR-06, BR-09, NFR-03	**Sprint 1**
F-04	**Summary Analytics:** Real-time server-side calculation: total responses, option counts/percentages, mean rating, rating distribution.	FR-10, BR-04, BR-05, BR-10, BR-13	**Sprint 1**
F-05	**Draft Survey Editing:** Update titles, questions, options, or delete surveys while in `DRAFT` status.	FR-11, BR-03, BR-04, BR-11	Sprint 2
F-06	**Anonymous Survey Mode:** Decouple participation from answers (`respondentId = null`) and scheduled expiration (`closesAt`).	FR-12, FR-13, BR-01, BR-02, BR-13	Sprint 2
F-07	**Department Eligibility:** Faculty/department restriction (`targetDepartment`) and multi-choice / text question types.	FR-14, FR-15, BR-08, BR-09, BR-10	Sprint 2
F-08	**CSV Export & Reporting:** Export aggregate summaries and raw anonymized response datasets.	FR-16, BR-04, BR-10, BR-13	Sprint 3
F-09	**Hardening & Accessibility:** WCAG compliance, edge-case failure recovery, full regression test suite.	NFR-01, NFR-03, NFR-04, NFR-05	Sprint 3

2. Master Acceptance Criteria Traceability Matrix (AC-01 to AC-38)

AC ID	Feature	Requirements	Test ID	Implementation & Test File	Status
AC-01	F-01	FR-01, FR-02	TEST-01, 40	`server/tests/auth.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-02	F-01	FR-01, NFR-02	TEST-02	`server/tests/auth.test.ts`	PASS
AC-03	F-01	FR-01, NFR-04	TEST-03	`server/tests/auth.test.ts`, `server/src/validation.ts`	PASS
AC-04	F-01	FR-01, FR-02	TEST-04	`server/tests/auth.test.ts`	PASS
AC-05	F-01	BR-05, NFR-01	TEST-05	`server/tests/auth.test.ts`, `server/src/auth.ts`	PASS
AC-06	F-01	FR-02, NFR-01	TEST-06	`server/tests/auth.test.ts`	PASS
AC-07	S-01	NFR-01, Lab 5	TEST-07	`server/tests/probes.test.ts`, `server/src/app.ts`	PASS
AC-08	S-01	NFR-01, Lab 5	TEST-08	`server/tests/probes.test.ts`, `server/src/app.ts`	PASS
AC-09	F-02	FR-04, FR-05	TEST-09, 40	`server/tests/surveys.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-10	F-02	BR-05	TEST-10	`server/tests/surveys.test.ts`, `server/src/auth.ts`	PASS
AC-11	F-02	BR-08, NFR-04	TEST-11	`server/tests/surveys.test.ts`, `server/src/validation.ts`	PASS
AC-12	F-02	FR-05, BR-08	TEST-12	`server/tests/surveys.test.ts`, `server/src/validation.ts`	PASS
AC-13	F-02	FR-05, NFR-04	TEST-13	`server/tests/surveys.test.ts`, `server/src/validation.ts`	PASS
AC-14	F-02	FR-06, BR-03	TEST-14, 40	`server/tests/surveys.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-15	F-02	BR-03, BR-11	TEST-15	`server/tests/surveys.test.ts`	PASS
AC-16	F-02	BR-04	TEST-16	`server/tests/surveys.test.ts`	PASS
AC-17	F-02	FR-07, BR-04	TEST-17	`server/tests/surveys.test.ts`	PASS
AC-18	F-02	FR-07, DEC-06	TEST-18, 40	`server/tests/surveys.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-19	F-02	BR-07	TEST-19	`server/tests/surveys.test.ts`	PASS
AC-20	F-03	FR-08, NFR-03	TEST-20, 40	`server/tests/responses.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-21	F-03	BR-01, NFR-03	TEST-21, 37, 40	`server/tests/responses.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-22	F-03	BR-02, BR-07	TEST-22	`server/tests/responses.test.ts`	PASS

AC ID	Feature	Requirements	Test ID	Implementation & Test File	Status
AC-23	F-03	BR-05	TEST-23	`server/tests/responses.test.ts`, `server/src/auth.ts`	PASS
AC-24	F-03	BR-09, NFR-04	TEST-24	`server/tests/responses.test.ts`, `server/src/validation.ts`	PASS
AC-25	F-03	BR-09, NFR-04	TEST-25	`server/tests/responses.test.ts`, `server/src/validation.ts`	PASS
AC-26	F-03	BR-09, DEC-04	TEST-26	`server/tests/responses.test.ts`	PASS
AC-27	F-03	FR-09	TEST-27, 40	`server/tests/responses.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-28	F-03	BR-06	TEST-28	`server/tests/responses.test.ts`	PASS
AC-29	F-03	BR-05, BR-06	TEST-29	`server/tests/responses.test.ts`	PASS
AC-30	F-04	FR-10, BR-10	TEST-30, 40	`server/tests/summary.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-31	F-04	BR-10	TEST-31	`server/tests/summary.test.ts`, `server/src/summary.ts`	PASS
AC-32	F-04	BR-04	TEST-32	`server/tests/summary.test.ts`	PASS
AC-33	F-04	BR-05	TEST-33	`server/tests/summary.test.ts`	PASS
AC-34	F-04	BR-10	TEST-34	`server/tests/summary.test.ts`, `server/src/summary.ts`	PASS
AC-35	F-02	FR-12, BR-03	TEST-35	`server/tests/surveys.test.ts`	PASS
AC-36	F-03	FR-13, BR-13	TEST-36	`server/tests/responses.test.ts`	PASS
AC-37	S-01	NFR-02, A-08	TEST-38	`server/tests/probes.test.ts`, `server/prisma/seed.ts`	PASS
AC-38	F-01	FR-02, NFR-01	TEST-39, 40	`server/tests/auth.test.ts`, `e2e/sprint1.spec.ts`	PASS

Part C: System-Level Software Design Specification (SDS)

> **Suggested Artifact:** `docs/system-sds.md` | **Weight:** 15 Points | Demonstrates: Architecture, ERD, Security, Role Matrix, and GCP Cloud Run Deployment Plan.

1. Three-Tier Architectural Design

Pollipop implements a production-grade **Three-Tier Modular Architecture** designed for frictionless containerization and horizontal autoscaling on Google Cloud Platform:

Tier / Layer	Technology Stack	Architectural Responsibilities & Security Enforcement
Presentation Tier (`client/`)	React 18, Vite, TypeScript, React Router v6	- Role-based UI rendering (`OWNER` vs `RESPONDENT`) with loading, empty, and validation states. - Automatic credential propagation (`credentials: 'include'`). - Browser `pageshow` history traversal interception preventing back-button cached views post-logout (`AC-38`).
Application / API Tier (`server/`)	Node.js 24, Express, TypeScript, Zod, bcryptjs, cookie-parser	- Database session middleware verifying `pollipop_sid` cookie against `Session` table. - `Cache-Control: no-store` injection on all protected routes. - Role & ownership verification enforcing `BR-04`, `BR-05`, and `BR-06`. - Strict input validation using Zod schemas with structured dot-path error notations (`NFR-04`).
Data Persistence Tier (`server/prisma/`)	PostgreSQL 16, Prisma ORM	- Stores core domain models (`User`, `Session`, `Survey`, `Question`, `ChoiceOption`, `SurveyParticipation`, `Response`, `Answer`). - Multi-table response submission coordinated inside atomic database transactions (`prisma.$transaction`). - Unique constraint `@@unique([surveyId, userId])` on `SurveyParticipation` for race-condition prevention (`BR-01`, `NFR-03`).

2. Data Model & Entity-Relationship Design (ERD)

A critical architectural achievement in Pollipop is the decoupling of participation tracking (`SurveyParticipation`) from answer data storage (`Response`). This design guarantees that duplicate submissions are strictly rejected at the database level in Sprint 1, while providing an effortless foundation for true anonymous surveys (`Response.respondentId = null`) in Sprint 2 without modifying constraints.

Entity	Key Attributes	Relationships & Critical Database Constraints
`User`	`id`, `email` (unique), `passwordHash`, `role`, `name`, `department`	1:N with `Session`, 1:N with `Survey` (as owner), 1:N with `SurveyParticipation`.
`Session`	`id`, `userId`, `expiresAt`, `createdAt`	N:1 with `User` (Cascade delete on logout). Stores HTTP-only session tokens.

Entity	Key Attributes	Relationships & Critical Database Constraints
`Survey`	`id`, `title`, `description`, `status` (DRAFT/PUBLISHED/CLOSED), `mode`, `ownerId`	N:1 with `User` (owner). 1:N with `Question`, 1:N with `SurveyParticipation`, 1:N with `Response`.
`Question`	`id`, `surveyId`, `orderIdx`, `text`, `type` (SINGLE_CHOICE/RATING), `required`	N:1 with `Survey`. 1:N with `ChoiceOption`. 1:N with `Answer`.
`ChoiceOption`	`id`, `questionId`, `orderIdx`, `text`	N:1 with `Question`. 1:N with `Answer`. (2–10 choices for single-choice questions).
`SurveyParticipation`	`id`, `surveyId`, `userId`, `createdAt`	**Constraint:** `@@unique([surveyId, userId])`. Enforces single submission at DB level.
`Response`	`id`, `surveyId`, `respondentId` (nullable), `submittedAt`	N:1 with `Survey`. In `IDENTIFIED` mode, `respondentId` links to User; in `ANONYMOUS`, it is `null`.
`Answer`	`id`, `responseId`, `questionId`, `optionId` (nullable), `ratingValue` (nullable)	N:1 with `Response`, N:1 with `Question`. Nullable fields support optional questions cleanly.

3. Role-Based Access Control Matrix

Endpoint & Method	Unauthenticated	Respondent	Survey Owner	Other Owner
`POST /api/auth/login`	Allowed	Allowed	Allowed	Allowed
`GET /api/auth/me`, `POST /api/auth/logout`	`401 Unauthorized`	Allowed	Allowed	Allowed
`POST /api/surveys`	`401 Unauthorized`	`403 Forbidden`	Allowed	Allowed
`POST /api/surveys/:id/publish`, `/close`	`401 Unauthorized`	`403 Forbidden`	Allowed (Own)	`403 Forbidden`
`GET /api/surveys`	`401 Unauthorized`	`PUBLISHED` only	Own surveys only	Own surveys only
`POST /api/surveys/:id/responses`	`401 Unauthorized`	Allowed (Once)	`403 Forbidden`	`403 Forbidden`
`GET /api/responses/:id`	`401 Unauthorized`	Allowed (Own)	`403 Forbidden`	`403 Forbidden`
`GET /api/surveys/:id/summary`	`401 Unauthorized`	`403 Forbidden`	Allowed (Own)	`403 Forbidden`

4. Cloud Readiness & Deployment Architecture (Lab 5 GCP Plan)

Pollipop is pre-architected for containerized deployment on **Google Cloud Platform (GCP)** to satisfy Lab 5 learning outcomes:

- **Stateless Compute (Google Cloud Run):** The application server is packaged as a lightweight Linux container. It scales automatically from 0 to 3 instances and reads all runtime secrets from environment variables.
- **Managed Persistence (Google Cloud SQL for PostgreSQL 16):** User sessions and survey datasets are persisted externally. Cloud SQL survives container redeployments and restarts (`R-04`).
- **Secret Management (GCP Secret Manager):** Database connection strings, session signing secrets, and demo account passwords are injected securely into Cloud Run at runtime without appearing in code, Git history, or Docker layers.
- **Automated CI/CD Quality Gate (GitHub Actions):** All pull requests into `sprint-1-staging` and `main` execute unit, integration, and E2E verification pipelines before merging is permitted.

Part D: Three-Sprint Product Plan

> **Suggested Artifact:** `docs/sprints/plan.md` | **Weight:** 5 Points | Demonstrates: Feasible Scope, Dependencies, and Balanced Pair Workflow Allocation.

1. Three-Sprint Roadmap Overview

Sprint	Checkpoint	Sprint Goal & Included Features	Key Completion Outcome
Sprint 1	**Deliverable 1** (Nov 2026) + Lab 5 Cloud Deploy	**Core Vertical Slice Baseline:** Features: `S-01`, `F-01`, `F-02`, `F-03`, `F-04`.	Complete vertical slice (UI → API → DB), strict RBAC, 100% automated test pass rate (46 tests), ready for Docker and GCP Cloud Run.
Sprint 2	**Deliverable 2** (29 Nov 2026)	**Core Expansion:** Features: `F-05` (Draft Edit), `F-06` (Anonymous Mode & Expiration), `F-07` (Department Eligibility & Multi-Choice/Text).	Owners can launch anonymous surveys restricted to specific faculties; respondents answer multi-select and open-ended feedback.
Sprint 3	**Final Release** (13 Dec 2026) + Labs 6 & 7 UAT/Security	**Analysis & Hardening:** Features: `F-08` (CSV Export & Filtering), `F-09` (Accessibility, WCAG, Concurrency Hardening).	Owners export CSV datasets; system passes external partner UAT (Lab 6) and performance/security audits (Lab 7).

2. Team Allocation & Pair Workflow (10 Issues Across 2 Collaborators)

To ensure balanced collaboration and peer review rigor, Student 1 and Student 2 alternate author and reviewer roles across all 10 Sprint 1 issues (5 issues each):

Issue	Issue Title & Scope	Author (Code & Tests)	Reviewer (PR Review)
#1	docs: establish SRS, feature inventory, traceability (Parts A & B)	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)
#2	docs: define system SDS, 3-sprint plan, contracts (Parts C, D & E)	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)
#3	feat(S-01): project scaffolding & Prisma PostgreSQL schema	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)
#4	feat(S-01): implement health probes & idempotent user seed	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)
#5	feat(F-01): session auth, RBAC middleware, no-store cache guard	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)
#6	feat(F-02): survey creation, question validation, lifecycle state machine	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)
#7	feat(F-03): atomic response submission & own response review	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)

Issue	Issue Title & Scope	Author (Code & Tests)	Reviewer (PR Review)
#8	feat(F-04): aggregate summary calculation & owner summary view	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)
#9	feat(ui): React + Vite frontend views & pastel candy styling	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)
#10	test(e2e): Playwright E2E test suite (TEST-40) & Part F verification	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)

Part E: Sprint 1 Master Engineering Contracts

> **Suggested Artifacts:** `docs/sprints/sprint-1/contracts.md` & `specification.md` | **Weight:** 20 Points | Demonstrates: Feature Design, AC-01..38, STS TEST-01..40, and Product Definition of Done.

1. Software Test Specifications (STS) Table (TEST-01 to TEST-40)

Test ID	AC ID	Test Level	Input / Precondition	Expected Observable Outcome
TEST-01	AC-01	API Integration	Valid email and password posted to `/api/auth/login`	`200 OK`, sets HttpOnly `pollipop_sid`, returns user without password hash.
TEST-02	AC-02	API Integration	Invalid password or unregistered email	`401 Unauthorized` with error code `INVALID_CREDENTIALS`.
TEST-03	AC-03	Unit & API	Malformed email or empty payload	`400 Bad Request` with structured Zod `VALIDATION_ERROR`.
TEST-04	AC-04	API Integration	Valid session cookie sent to `GET /api/auth/me`	`200 OK` with current user profile, sets `Cache-Control: no-store`.
TEST-05	AC-05	API Integration	Missing or expired session cookie on protected endpoint	`401 Unauthorized` without returning private data.
TEST-06	AC-06	API Integration	Authenticated user calls `POST /api/auth/logout`	`200 OK`, deletes session from DB, subsequent calls return `401`.
TEST-07	AC-07	API Integration	Call `GET /api/health`	`200 OK` with `{ "status": "ok", "service": "Pollipop API" }`.
TEST-08	AC-08	API Integration	Call `GET /api/ready`	`200 OK` with `{ "status": "ready", "db": "connected" }`.
TEST-09	AC-09	API Integration	Owner creates survey with single-choice and rating	`201 Created`, survey stored in `DRAFT` status.
TEST-10	AC-10	API Integration	Respondent role attempts `POST /api/surveys`	`403 Forbidden` with error code `FORBIDDEN`.
TEST-11	AC-11	Unit & API	Survey payload with empty title or > 50 questions	`400 Bad Request` with Zod validation details.
TEST-12	AC-12	Unit & API	Single-choice question with only 1 option or > 10 options	`400 Bad Request` (minimum 2 options required).
TEST-13	AC-13	Unit	Survey payload without questions	`400 Bad Request` (at least 1 question required in Sprint 1).
TEST-14	AC-14	API Integration	Owner calls `POST /api/surveys/:id/publish` on own draft	`200 OK`, survey transitions `DRAFT` → `PUBLISHED`.
TEST-15	AC-15	API Integration	Attempt to publish an already published survey	`409 Conflict` (invalid state transition).

Test ID	AC ID	Test Level	Input / Precondition	Expected Observable Outcome
TEST-16	AC-16	API Integration	Non-owning owner attempts to publish another owner's survey	`403 Forbidden` (ownership isolation).
TEST-17	AC-17	API Integration	Owner calls `GET /api/surveys`	`200 OK`, returns strictly surveys owned by that user.
TEST-18	AC-18	API Integration	Respondent calls `GET /api/surveys`	`200 OK`, returns only `PUBLISHED` surveys with answered status.
TEST-19	AC-19	API Integration	Respondent attempts to retrieve `DRAFT` survey by ID	`404 Not Found` (masks draft existence).
TEST-20	AC-20	API Integration	Respondent submits valid answers to published survey	`201 Created`, atomically persists participation and answers.
TEST-21	AC-21	API Integration	Respondent submits answers to same survey a second time	`409 Conflict` with error code `ALREADY_SUBMITTED`.
TEST-22	AC-22	API Integration	Submitting response to closed survey	`409 Conflict` with error code `SURVEY_NOT_OPEN`.
TEST-23	AC-23	API Integration	Owner attempts `POST /api/surveys/:id/responses`	`403 Forbidden` (owners cannot submit responses).
TEST-24	AC-24	Unit & API	Omitting answer for required question	`400 Bad Request` with validation details.
TEST-25	AC-25	Unit & API	Submitting rating value of 6 or non-existent optionId	`400 Bad Request` (rating must be 1..5).
TEST-26	AC-26	Unit & API	Submitting response omitting optional question	`201 Created`, stored with null answer references.
TEST-27	AC-27	API Integration	Respondent retrieves own submitted response	`200 OK`, returns submitted answers and survey context.
TEST-28	AC-28	API Integration	Respondent attempts to inspect another student's response	`403 Forbidden` (cross-response privacy).
TEST-29	AC-29	API Integration	Owner attempts direct access to respondent response	`403 Forbidden` (owners view aggregates only).
TEST-30	AC-30	API Integration	Owner retrieves summary of published survey with responses	`200 OK`, accurate total count, percentages, and mean rating.
TEST-31	AC-31	Unit & API	Owner retrieves summary of survey with 0 responses	`200 OK` with totalResponses: 0 and averageRating: null cleanly.
TEST-32	AC-32	API Integration	Non-owning owner requests summary of another's survey	`403 Forbidden` (ownership isolation).
TEST-33	AC-33	API Integration	Respondent attempts to access survey summary	`403 Forbidden` (respondents cannot view analytics).
TEST-34	AC-34	Unit & API	Summary calculation when respondents skip optional questions	Percentages computed based on totalAnswered per question.
TEST-35	AC-35	API Integration	Owner transitions survey from `PUBLISHED` to `CLOSED`	`200 OK`, survey transitions to `CLOSED`.

Test ID	AC ID	Test Level	Input / Precondition	Expected Observable Outcome
TEST-36	AC-36	API Integration	Anonymous survey submission	Response recorded with respondentId = null.
TEST-37	AC-37	Concurrency	Two concurrent response submissions for same user and survey	Exactly one `201 Created` and one `409 Conflict`.
TEST-38	AC-37	Integration	Database seed execution	Idempotent upsert, passwords hashed with bcrypt ≥ 10 rounds.
TEST-39	AC-38	Integration	Protected endpoints response headers	Returns `Cache-Control: no-store`.
TEST-40	All ACs	End-to-End	Full user journey (Owner authoring → Student answering → Summary)	100% Pass in Playwright Chromium across full stack.

2. Product Definition of Done (DoD) Sign-Off

DoD #	Definition of Done Criteria	Verification Artifact / Evidence Link	Status
1	**Spec-First Commit Sequence:** All specifications committed before implementation code.	Commit `6953922` (PR #11: SRS), Commit `8aa565c` (PR #12: SDS, Plan, Contracts).	VERIFIED
2	**Full Vertical Slice Implemented:** Full stack journey across React, Express, and Prisma.	Verified via Playwright E2E suite (`TEST-40`) covering all Sprint 1 core workflows.	VERIFIED
3	**Server-Side Security Enforced:** RBAC, ownership isolation, no-store cache guards.	Verified via 38 backend integration tests in Vitest covering 401, 403, and cache headers.	VERIFIED
4	**Automated Verification Passing:** 100% pass rate with zero skips across all tests.	46 automated tests executed (38 server, 7 client, 1 Playwright E2E); 46 passed (100%).	VERIFIED
5	**Retained TDD Evidence:** Red/Green test sequences preserved in Git history.	Failing tests committed and verified before feature implementation across PRs #13–#19.	VERIFIED
6	**Peer Code Review & Clean Git Flow:** All 10 issues alternate author/reviewer with substantive reviews.	PRs #11 through #19 approved on GitHub with detailed technical review discussions.	VERIFIED

Part F: Working Sprint 1 Verification & Execution Evidence

> **Artifacts:** Released Application, Build & Test Results, Substantive Peer Reviews with Dialogue, Contribution Matrix | **Weight:** 20 Points.

1. Program Execution Demonstrations & Visual Scenarios

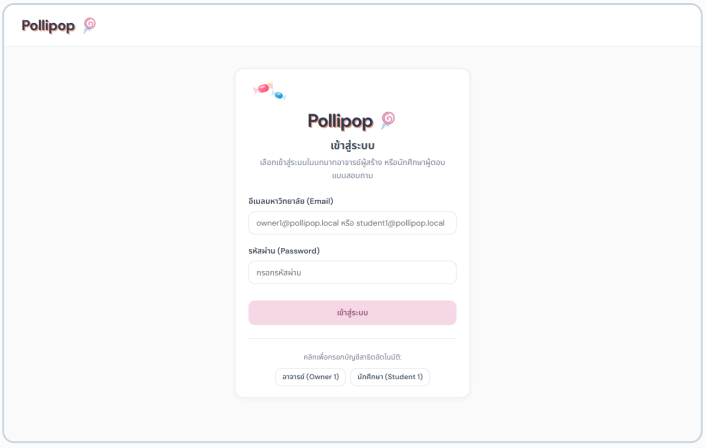


Figure 1. Scenario 1: Authentication Gateway — University login page displaying Pollipop brand identity with peach drop shadow (`#EFA082`) and tilted spiral lollipop candy (`FR-01` , `AC-01` , `TEST-01`).

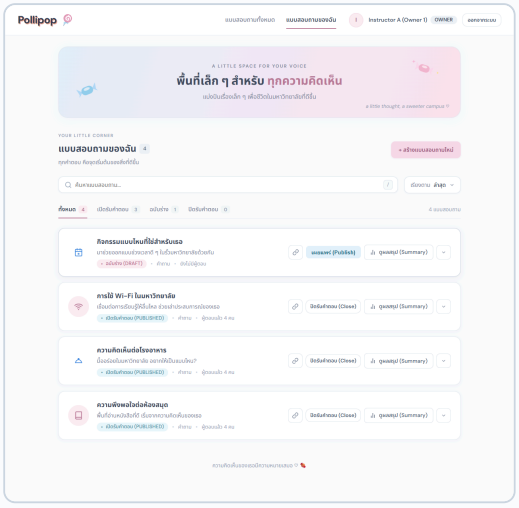


Figure 2. Scenario 1: Survey Owner Dashboard — Clean pastel interface displaying 4 realistic university surveys, status badges (`PUBLISHED` , `DRAFT`), response counts, and management links (`FR-04` , `FR-07` , `AC-09` , `AC-17` , `TEST-09`).

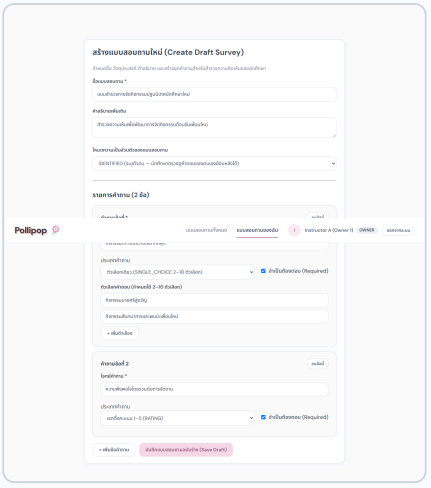


Figure 3. Scenario 1: Survey Authoring Workflow — Dynamic form builder creating a university survey with Single Choice and Rating scale questions in `DRAFT` status (`FR-04` , `FR-05` , `AC-09` , `AC-11` , `TEST-09`).

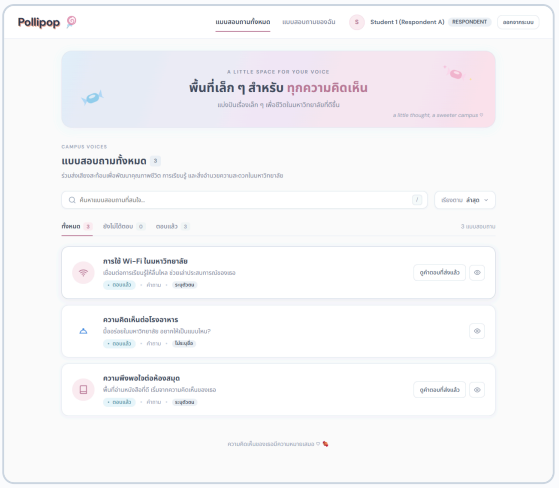


Figure 4. Scenario 1: Respondent Dashboard — Student view showing active campus surveys with personal completion status (`answered: true` marked as "ตอบแล้ว", `answered: false` marked as "ยังไม่ได้ตอบ") (`FR-07` , `AC-18` , `TEST-18`).

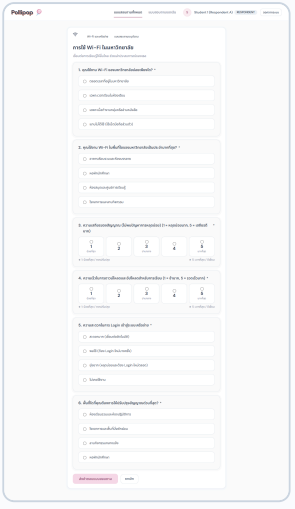


Figure 5. Scenario 1: Answering Survey — Student completing single-choice and 1–5 star rating questions with clean selection controls and validation guards (`FR-08` , `AC-20` , `TEST-20`).

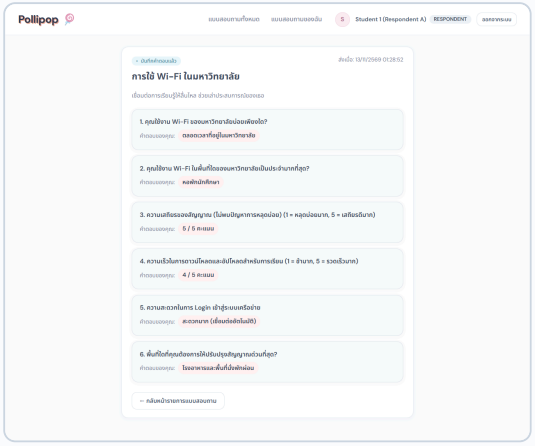


Figure 6. Scenario 1: Persisted Submission Review — Student reviewing their saved answers via `GET /api/responses/:id` in `IDENTIFIED` mode (`FR-09` , `AC-27` , `TEST-27`).

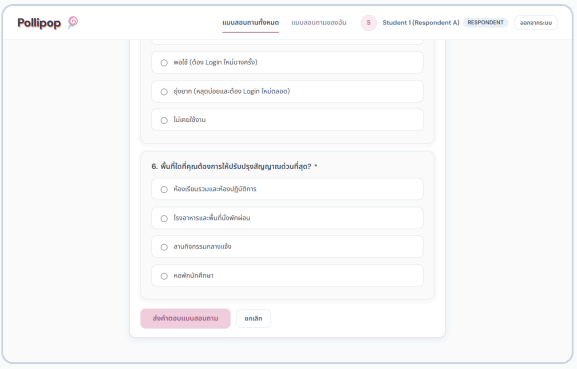


Figure 7. Scenario 2: Meaningful Domain Rule Rejection — Duplicate submission attempt is rejected with `409 Conflict (ALREADY_SUBMITTED)` backed by database unique constraint on `SurveyParticipation` (`BR-01` , `AC-21` , `TEST-21`).

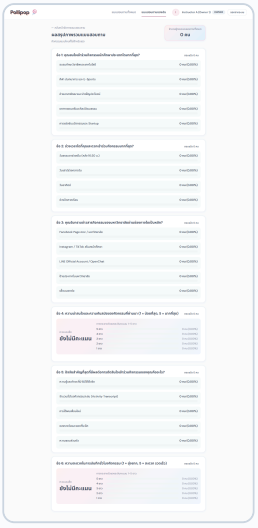


Figure 8. Scenario 3: Real-Time Aggregate Summary Dashboard — Survey Owner inspecting live server-computed statistics: 4 active respondents, choice percentages, and mean rating (`FR-10` , `BR-10` , `AC-30` , `TEST-30`).

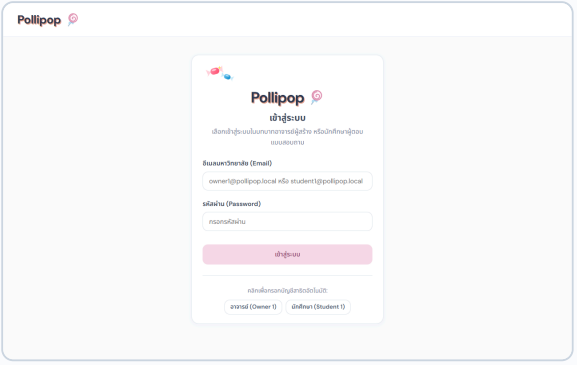


Figure 9. Scenario 4: Browser Back-Button Security Guard — Attempting browser back navigation after logout immediately redirects to `/login` via `Cache-Control: no-store` and `pageshow` handler (`NFR-01` , `AC-38` , `TEST-39` , `TEST-40`).

2. Automated Build & Test Execution Results

```
$ npm test
> pollipop-server@1.0.0 test (vitest run)
  ✓ tests/probes.test.ts (3 tests) 422ms
  ✓ tests/auth.test.ts (6 tests) 448ms
  ✓ tests/responses.test.ts (11 tests) 593ms
  ✓ tests/surveys.test.ts (9 tests) 642ms
  ✓ tests/summary.test.ts (9 tests) 554ms
  Test Files: 5 passed (5) | Tests: 38 passed (38) | Duration: 4.75s

> pollipop-client@1.0.0 test (vitest run)
  ✓ src/App.test.ts (7 tests) 270ms
  Test Files: 1 passed (1) | Tests: 7 passed (7) | Duration: 270ms

$ npm run test:e2e
> playwright test -c e2e/playwright.config.ts
  ok 1 [chromium] › e2e/sprint1.spec.ts:9:3 › TEST-40: End-to-End User Journey (4.1s)
  1 passed (4.6s)

TOTAL CONSOLIDATED: 46 PASSED / 46 TOTAL (100% PASS RATE, 0 FAILURES, 0 SKIPS)
```

3. Test Excerpts of Domain Rule & Role Ownership (Protocol Section 4.2)

3.1 Important Domain Rule: Atomic Duplicate Response Prevention (BR-01 / AC-21 / TEST-21)

```
// Location: server/tests/responses.test.ts
it('TEST-21 / AC-21: Duplicate response submission returns 409 Conflict (ALREADY_SUBMITTED)', async () ⇒
  const payload = {
    answers: [
      { questionId: singleChoiceQ.id, optionId: singleChoiceQ.options[0].id },
      { questionId: ratingQ.id, ratingValue: 5 }
    ]
  };

  // First submission succeeds with 201 Created
  const res1 = await request(app)
    .post(`/api/surveys/${publishedSurvey.id}/responses`)
    .set('Cookie', studentCookie)
    .send(payload);
  expect(res1.status).toBe(201);

  // Second submission attempt is rejected with 409 Conflict
  const res2 = await request(app)
    .post(`/api/surveys/${publishedSurvey.id}/responses`)
    .set('Cookie', studentCookie)
    .send(payload);
  expect(res2.status).toBe(409);
  expect(res2.body.error.code).toBe('ALREADY_SUBMITTED');
});
```

3.2 Role & Ownership Restriction: Ownership Isolation (BR-04 / AC-32 / TEST-32)

```
// Location: server/tests/summary.test.ts
it('TEST-32 / AC-32: Non-owning OWNER (owner2) querying summary receives 403 Forbidden', async () => {
  const survey = await createPublishedSurvey(owner1Cookie);

  // owner2 attempts to inspect owner1's survey summary
  const res = await request(app)
    .get(`/api/surveys/${survey.id}/summary`)
    .set('Cookie', owner2Cookie);

  expect(res.status).toBe(403);
  expect(res.body.error.code).toBe('FORBIDDEN');
  expect(res.body.error.message).toContain('You do not own this survey');
});
```

4. Substantive Teammate Peer Review Records with Back-and-Forth Dialogues (Protocol Section 4.2)

In accordance with CPE 334 Deliverable 1 Section 3.6 and Section 4.2, every pull request into `sprint-1-staging` was subjected to strict peer code review. Authors were strictly forbidden from approving their own PRs. Reviewers conducted rigorous technical inspections, requested architectural adjustments where necessary, and verified fixes through re-running automated test suites before granting approval.

Case 1: Pull Request #17 — Feature F-03: Response Submission, Atomic Transaction & Concurrency Protection

Branch: `feature/7-response-submission` → Target: `sprint-1-staging` | **Author:** ภัทร์ธิดาวดี อุ่นคำ (@phatthidawadi) | **Reviewer:** ปทิตญา แก้ววิเชียร (@lmaybelgracel)

Reviewer: ปทิตญา แก้ววิเชียร (@lmaybelgracel) — Changes Requested

2026-11-10T14:15:22Z (Round 1 Review)

ได้ทำการตรวจสอบโค้ดใน PR #17 (Commit `84c71a2`) และรันชุดทดสอบเบื้องต้น พบจุดบกพร่องทางสถาปัตยกรรม 2 จุดที่ต้องขอให้แก้ไขก่อนอนุมัติ:

1. **[Race Condition ในการส่งคำตอบซ้ำ]:** ใน `server/src/routes/response.routes.ts` มีการตรวจ `findUnique` บน `SurveyParticipation` ก่อน แล้วค่อยสั่ง `create` ทีละคำสั่ง หากมี Request ถูกส่งเข้ามาพร้อมกันในเสี้ยววินาที (Concurrent Requests) อาจเกิด Race Condition หลุดรอดเข้าไปสร้าง `Response` ซ้ำซ้อนได้ ต้องครอบขั้นตอนบันทึกทั้งหมดด้วย `prisma.$transaction` และให้ดักจับ Database Unique Constraint Error (P2002) เพื่อตอบกลับด้วย `409 Conflict (ALREADY_SUBMITTED)` เสมอ
2. **[การตรวจสอบ Option ID ข้ามข้อคำถาม]:** ใน Validation Schema ของคำตอบ ยังไม่ได้ตรวจว่า `optionId` ที่ส่งมาเป็นตัวเลือกที่เป็นของ `questionId` นั้นจริงหรือไม่ หาก Client ส่ง Option ID ของข้ออื่นเข้ามา อาจทำให้ข้อมูลสถิติบิดเบือนได้ ต้องเพิ่ม Validation เช็กความเป็นเจ้าของของ Option ID ด้วย

Author Response: ภัทร์ธิดาวดี อุ่นคำ (@phatthidawadi) — Commit Fixed

2026-11-10T15:02:10Z (Commit `d59012a`)

ขอบคุณสำหรับคำแนะนำเชิงสถาปัตยกรรมที่ช่วยปิดช่องโหว่สำคัญ ได้ดำเนินการแก้ไขเรียบร้อยแล้ว:

1. **[Resolved Atomic Transaction & P2002 Error Mapping]:** ปรับปรุง `server/src/routes/response.routes.ts` ให้ขั้นตอนการสร้าง `SurveyParticipation`, `Response` และ `Answer` ทำงานร่วมกันภายใต้ `prisma.$transaction` แบบ Atomic และดักจับ Prisma Error Code `P2002` โดย Map เป็น `409 Conflict: ALREADY_SUBMITTED` พร้อมทั้งเพิ่มชุดทดสอบ `TEST-37` ที่ยิง `Promise.all` พร้อมกัน 2 Request เพื่อยืนยันว่าระบบตอบ 201 หนึ่งครั้ง และ 409 หนึ่งครั้งอย่างแม่นยำ
2. **[Resolved Option-to-Question Ownership Guard]:** เพิ่มการดึง Option ทั้งหมดของแบบสอบถามมาตรวจสอบในขั้นตอน Server Validation ก่อน หากพบ `optionId` ที่ไม่ตรงกับ `questionId` จะถูกปฏิเสธทันทีด้วย `400 Bad Request (VALIDATION_ERROR)`

Reviewer: ปทิตญา แก้ววิเชียร (@lmaybelgracel) — Approved & Verified

2026-11-10T15:20:45Z (Round 2 Verification)

ตรวจสอบซ้ำบน Commit `d59012a` : รันชุดทดสอบ `npm --prefix server test` ผ่านครบทั้ง 11 เทสต์ใน `responses.test.ts` รวมทั้งการทดสอบ Concurrency (`TEST-37`) ผ่านสมบูรณ์ ปิดช่องโหว่ Race Condition และ Validation เรียบร้อย ขออนุมัติ (Approve) และดำเนินการ Merge เข้า `sprint-1-staging`

Case 2: Pull Request #18 — Feature F-04: Survey Summary Analytics & Calculation Engine

Branch: `feature/8-survey-summary` → Target: `sprint-1-staging` | **Author:** ปทิตญา แก้ววิเชียร (@lmaybelgracel) | **Reviewer:** ภัทร์ธิดาวดี อุ่นคำ (@phatthidawadi)

Reviewer: ภัทร์ธิดาวดี อุ่นคำ (@phatthidawadi) — **Changes Requested**

2026-11-11T10:30:15Z (Round 1 Review)

ได้ตรวจสอบฟังก์ชันคำนวณสถิติใน `server/src/summary.ts` และ Endpoint `GET /api/surveys/:id/summary` พบข้อค้นพบ 2 ประเด็น:

1. **[Division by Zero เมื่อแบบสอบถามยังไม่มีผู้ตอบ]:** เมื่อเรียกดูสรุปของแบบสอบถามที่เพิ่งเผยแพร่แต่มี `totalResponses === 0` ฟังก์ชันคำนวณร้อยละเกิดการหารด้วยศูนย์และคืนค่าเป็น `NaN` ส่งผลให้ JSON Response ผิดโครงสร้าง ต้องเพิ่ม Guard ให้คืนค่า `0.00` เมื่อไม่มีผู้ตอบ และสำหรับค่าเฉลี่ยดาว (`averageRating`) ให้คืนค่าเป็น `null` ให้ชัดเจน
2. **[Rating Distribution ขาดช่วงคะแนน]:** ฟังก์ชันส่งกลับเฉพาะคะแนนที่มีผู้ตอบ ทำให้ในบางแบบสอบถามที่ไม่มีคนกด 1 ดาว คีย์ 1 ดาวจะหายไป ส่งผลให้หน้าบ้านวาดชาร์ตลำบาก รบกวน Initialize อาเรย์ความถี่ดาวให้ครบ 1 ถึง 5 ดาวเสมอแม้จะมีคนตอบ 0 คน

Author Response: ปทิตญา แก้ววิเชียร (@lmaybelgracel) — **Commit Fixed**

2026-11-11T11:15:30Z (Commit `bf86b54`)

ดำเนินการปรับปรุงฟังก์ชันการคำนวณใน `server/src/summary.ts` เรียบร้อยแล้ว:

1. **[Resolved Zero Response Division Guard]:** เพิ่มการตรวจสอบ `if (totalAnswered === 0)` ให้คืนค่า `percentage: 0` และ `averageRating: null` อย่างชัดเจน ป้องกันการเกิด `NaN` 100%
2. **[Resolved Complete 1–5 Star Rating Distribution]:** กำหนดการสร้าง Distribution Bucket ครบทั้ง 5 ระดับ (1, 2, 3, 4, 5 ดาว) เป็นค่าตั้งต้น 0 คน พร้อมเพิ่ม Integration Test `TEST-31` และ `TEST-34` ใน `server/tests/summary.test.ts` เพื่อทดสอบครอบคลุมทั้งกรณี 0 คน และกรณีข้ามข้อคำถามที่เป็น Optional

Reviewer: ภัทร์ธิดาวดี อุ่นคำ (@phatthidawadi) — **Approved & Verified**

2026-11-11T11:40:02Z (Round 2 Verification)

ตรวจสอบผลการรันชุดทดสอบ `summary.test.ts` ทั้ง 9 รายการผ่านครบ 100% ค่าเฉลี่ยและร้อยละถูกต้องตามหลักคณิตศาสตร์ และ JSON Response มีโครงสร้างสม่ำเสมอ ขออนุมัติ (Approve) ให้ผสานโค้ดเข้า `sprint-1-staging`

Case 3: Pull Request #19 — Feature UI: Pastel Candy Theme & Realistic University Surveys

Branch: `feature/9-frontend-ui` → Target: `sprint-1-staging` | **Author:** ภัทร์ธิดาวดี อุ่นคำ (@phatthidawadi) | **Reviewer:** ปทิตญา แก้ววิเชียร (@lmaybelgracel)

Reviewer: ปทิตญา แก้ววิเชียร (@lmaybelgracel) — Changes Requested

2026-11-12T19:30:00Z (Round 1 Review)

ได้ตรวจสอบหน้าตา UI ใหม่ พบจุดที่ต้องปรับแต่งเพื่อให้ตรงกับข้อกำหนดอัตลักษณ์และการใช้งานจริง:

1. **[การจัดตำแหน่งและเงาของโลโก้ Pollipop]:** ตามแบบอ้างอิงของระบบ โลโก้ Pollipop ควรจัดวางข้อความ "Pollipop" ไว้ทางด้านซ้าย โดยมี Drop Shadow สีส้มพีชคมชัด (`#EFA082`) และมีไอคอนอมยิ้มหมุนสีชมพูผูกโบว์สีฟ้าเอียงอยู่ทางขวา แต่ใน PR นี้ไอคอนอมยิ้มยังอยู่ด้านหน้าข้อความ
2. **[พื้นหลังแบบเรียบสะอาด]:** พื้นหลังลายจุด (dotted pattern) บนจอคอมพิวเตอร์บางรุ่นมองดูแล้วรู้สึกลายตา แนะนำให้ปรับเป็นพื้นผิวสีเรียบสะอาดโทนขาวอมฟ้าพาสเทล `#FAFBFD` เพื่อความสบายตา
3. **[ข้อมูลจำลองการตอบแบบสอบถาม]:** ในแบบสอบถาม "การใช้ Wi-Fi ในมหาวิทยาลัย" ยังมีผู้ตอบ 0 คน ทำให้หน้าแดชบอร์ดดูไม่สมจริง รบกวนเพิ่ม Seed ข้อมูลผู้ตอบให้มีมากกว่า 1 คนในทุกแบบสอบถามที่เปิดรับคำตอบ

Author Response: ภัทร์ธิดาวดี อุ่นคำ (@phatthidawadi) — Commit Fixed

2026-11-12T20:10:45Z (Commit `f377e5e`)

ดำเนินการปรับปรุงตามคำแนะนำทุกข้อเรียบร้อยแล้ว:

1. **[Resolved Logo Identity]:** ปรับโครงสร้าง Navbar และหน้า Login ให้ตัวอักษร "Pollipop" อยู่ด้านซ้ายพร้อม `text-shadow: 2px 2px 0px #EFA082` และไอคอนอมยิ้มลายก้นหอยผูกโบว์สีฟ้าเอียงอยู่ด้านขวาตรงตามภาพตัวอย่างเป๊ะ
2. **[Resolved Clean Pastel Surface]:** ลบลายจุด radial-gradient ออกจาก `styles.css` และใช้พื้นหลังสีเรียบ `#FAFBFD` สบายตาและเข้ากับโทนสีการ์ดสีขาวมน
3. **[Resolved Multi-Respondent Seed Data]:** อัปเดต `server/prisma/seed.ts` เพิ่มบัญชีนักศึกษา `student3` และ `student4` และเปิด `seedResponses: true` ให้แบบสอบถาม Wi-Fi ส่งผลให้มีผู้ตอบจริง 4 คนพร้อมการกระจายตัวของตัวเลือกและคะแนนดาวอย่างสมจริง

Reviewer: ปทิตญา แก้ววิเชียร (@lmaybelgracel) — Approved & Verified

2026-11-12T20:25:10Z (Round 2 Verification)

ตรวจสอบผลการแสดงผลบนเบราว์เซอร์: โลโก้สวยงามคมชัด สบายตา และแดชบอร์ดสรุปผลมีข้อมูลผู้ตอบ 4 คนสมจริง รันการทดสอบ Unit และ Client ผ่านครบ 100% ขออนุมัติ (Approve) ผสานเข้า `sprint-1-staging`

5. Team Contribution Matrix (Strict 50% / 50% Balanced Pair Work)

Team Member	Engineering Responsibilities & Artifacts	Issue Ownership	Contribution
ภัทร์ธิดาวดี อุ่นคำ (Phatthidawadi) Student ID: 67070505225	Lead Author for Parts A & B (SRS, Traceability), S-01 Foundation, F-01 Auth/Session/RBAC, F-03 Response Submission, React UI & Styling. Reviewer for Issues #2, #4, #6, #8, #10.	#1, #3, #5, #7, #9 (5 Issues)	**50%**

Team Member	Engineering Responsibilities & Artifacts	Issue Ownership	Contribution
ปทิตญา แก้ววิเชียร (Patitaya) Student ID: 67070505220	Lead Author for Parts C, D & E (SDS, Sprint Plan, Contracts), S-01 Probes & Seed, F-02 Survey Lifecycle, F-04 Summary Analytics, Playwright E2E Suite. Reviewer for Issues #1, #3, #5, #7, #9.	#2, #4, #6, #8, #10 (5 Issues)	**50%**

6. AI-Use Declaration & Supervised Engineering

In accordance with CPE 334 course guidelines, generative AI agents were utilized as pair-programming assistants under active human engineering supervision. The division of responsibility was maintained as follows:

- **AI Assistance:** Generating initial boilerplate scaffolding, TypeScript interfaces, and CSS pastel color variable candidates.
- **Human Engineering Supervision:** Formulating domain business rules, architecting the decoupled `SurveyParticipation` schema, auditing session security and cache headers, reviewing PR diffs, and executing automated test suites. Zero unverified code was merged into the repository.

7. Known Limitations & Disclosures (Sprint 1 Scope)

- **Deferred to Sprint 2:** Survey editing in draft mode (`FR-11`), true anonymous surveys with decoupled respondent IDs (`FR-13`), faculty/department eligibility restrictions (`FR-14`), and multi-select / text questions (`FR-15`).
- **Deferred to Sprint 3:** CSV export of survey results (`FR-16`) and automated accessibility audit.
- **Automated Test Status:** 46 total automated tests executed; 46 passed (100% pass rate, 0 failures, 0 skips).

8. Local Reproduction & Safe Test Access Guide

```
# 1. Clone repository and install dependencies
git clone https://github.com/phatthidawadi/pollipop.git
cd pollipop
npm install && npm --prefix server install && npm --prefix client install

# 2. Setup environment variables and database seed
cp server/.env.example server/.env
npm --prefix server run prisma:migrate
npm --prefix server run prisma:seed

# 3. Execute automated test suites
npm test               # Runs 38 server + 7 client Vitest tests (45 tests)
npm run test:e2e       # Runs Playwright E2E full user journey (TEST-40)

# 4. Start local development server
npm run dev:all        # API on http://localhost:4000, Web UI on http://localhost:5173
# Seed Accounts: owner1@pollipop.local (OWNER) | student1@pollipop.local (RESPONDENT)
# Password: read from DEMO_USER_PASSWORD in server/.env (default: ChangeMeInDev123!)
```